



RNA-Seq analysis on Tufts HPC

Part1 : HPC, Nextflow and nf-core

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Agenda

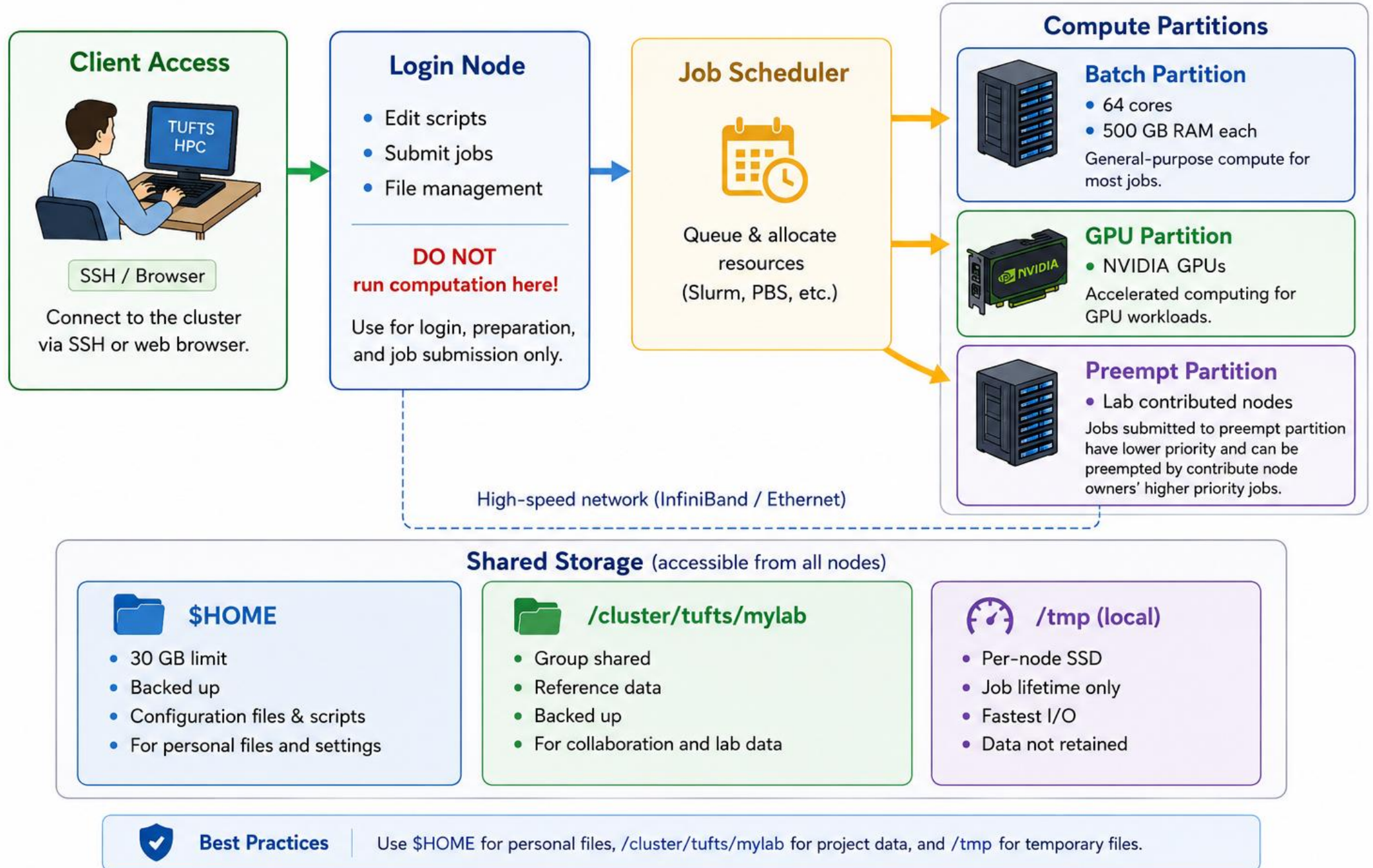
- Architecture of Tufts HPC
- Walk through Tufts Open OnDemand platform
- Connect to Tufts HPC and transfer files
- Submit, monitor, and cancel a simple SLURM job
- Introduction to Nextflow & nf-core
- RNA-Seq analysis with nf-core rnaseq pipeline on Tufts Open OnDemand



Part1 - Intro to Tufts HPC



Tufts HPC Cluster Architecture



The Key Components

1. Login node:

- Where you SSH in. Shared by all users. Never run analysis here.

2. Scheduler (slurm):

- Accepts resource requests, queues jobs, allocates compute nodes

3. Compute nodes:

- Many CPUs, lots of RAM. Allocated exclusively while your job runs.

4. Shared storage:

- Same filesystem visible from all nodes. Copy data once, use from anywhere.
- **These four components appear on every HPC system.**



Connecting to Tufts HPC

Tool	What it is	When to use it
SSH	Encrypted terminal connection to the cluster	Login cluster, edit files and submit jobs
rsync	copy and synchronize files/folders between locations.	Transfer or sync many files, large files, or interrupted transfers between cluster and your computer.
sftp	Interactive file transfer over SSH	Transfer files interactively between your computer and cluster.



Two Ways to Access Tufts HPC

- **OnDemand Web Portal:**
 - Browser-based — good for Jupyter, RStudio, exploring
 - URL: `ondemand-prod.pax.tufts.edu`
- **SSH Terminal:**
 - Direct command-line access from your laptop
 - Required for SLURM job submission and scripting
 - `ssh your_utln@login-prod.pax.tufts.edu`
- **Off campus: connect to Tufts VPN first.**



What is SSH?

- **SSH = Secure Shell**
 - Encrypted terminal connection to a remote computer
 - Already installed on Mac and Linux
 - Windows: Git Bash, WSL, or PowerShell
- **Tufts HPC login address: `login-prod.pax.tufts.edu`**



Connecting via SSH

```
# From your laptop terminal:  
$ ssh your_utln@login-prod.pax.tufts.edu  
  
# First time: confirm the host fingerprint – type 'yes'  
# Enter your Tufts password + 2FA  
  
# Your prompt changes to:  
[your_utln@login-prod-01 ~]$  
  
# 'login-prod' in the prompt = you are on the LOGIN NODE  
  
# Check your location:  
$ pwd      # /cluster/home/your_utln
```



The Login Node Rule

- You land on the LOGIN NODE when you SSH in.
- Login nodes are SHARED by everyone connected.
- OK on login nodes:
 - Edit files, submit jobs, check job status, light commands

- **Never run jobs or heavy commands on login nodes**

If you violate this, you do not just slow yourself down. You slow down everyone else on that login node. Admins will kill your job and warn you.

All real computation → SLURM → compute nodes.



Transferring Projects: rsync

- For whole project directories, use rsync.
- **Why rsync over SFTP?**
 - Transfers entire directory trees
 - Shows progress and can RESUME interrupted transfers
 - Smart: only copies files that have changed
- **Run rsync from your LAPTOP, not from inside SSH.**
- **rsync works over SSH — same security, better for bulk transfers.**



rsync: Upload and Download

```
# Upload a project to the cluster (run from your laptop):  
$ rsync -avzP ~/bio196/project/ \  
  your_utln@login-prod.pax.tufts.edu:/cluster/home/your_utln/project/  
  
# Download results from cluster to laptop:  
$ rsync -avzP \  
  your_utln@login-  
prod.pax.tufts.edu:/cluster/home/your_utln/project/results/ \  
  ~/bio196/project/results/  
  
# Flags:  -a archive  -v verbose  -z compress  -P progress+resume  
  
# TRAILING SLASH matters:  
#  project/    ->  copies CONTENTS  
#  project     ->  copies the folder itself (creates nested dir)
```



Data transfer with Globus

<https://access.tufts.edu/globus>

<https://rtguides.it.tufts.edu/storage/globus/index.html>



The screenshot shows the TTS Research Technology Guides website. The top navigation bar includes links for High-Performance Computing, Research Storage (selected), Bioinformatics, and More. A search bar and social media icons are also present. The left sidebar lists various guides, with 'Globus' selected. The main content area is titled 'Globus' and contains a summary of the guide, a 'What is Globus?' section, and a 'Key Concept: endpoint' section. The right sidebar provides additional links like 'Suggest Edits', 'Show Source', and 'Report Error'.

TTS Research Technology Guides High-Performance Computing **Research Storage** Bioinformatics More

Search [x] [K]

[Collapse Sidebar](#)

Research Storage

- Available Storage
- Research Computing Resource Manager
- Globus**
- Globus Account Setup
- Remote Endpoint - Tufts Globus Collections
- Local Endpoint - Globus Connect Personal
- Transfer Files between Two Endpoints
- Examples and Tutorials
- Help with Storage
- Research Storage Request

Globus

This step-by-step guide will show you how to log into Globus and use it to transfer files reliably and securely. You will become familiar with basic Globus concepts and interfaces, and begin to experience how Globus can help you spend more time on your research and less time on data management.

What is Globus?

[Globus](#) is research cyberinfrastructure, developed and operated as a not-for-profit service by the University of Chicago.

Globus provides a secure, unified interface to your research data. Use Globus to 'fire and forget' high-performance data transfers between systems within and across organizations.

Key Concept: *endpoint*

An endpoint is a server that hosts collections. If you want to be able to access, share, transfer, or manage data using Globus, the first step is to create an endpoint on the system where the data is (or will be) stored.

[Globus Connect](#) is used to create endpoints. An endpoint can be a laptop, a personal desktop system, a laboratory server, a campus data storage service, a cloud service, or an HPC cluster. As explained below, it's easy to set up your own Globus endpoint on a laptop or other personal system using Globus Connect Personal. Administrators of shared services (like campus storage servers) can set up multi-user endpoints using Globus Connect Server. You can use endpoints set up by others as long as you're authorized by the endpoint administrator or by a collection manager.

Globus

- What is Globus?
- Key Concept: *endpoint*
- [Suggest Edits](#)
- [Show Source](#)
- [Report Error](#)

What is a Job?

- A job contains two things:
- Resource request
 - partition
 - CPUs
 - GPUs (optional)
 - memory
 - runtime
- Commands to execute



Common resource and job options

Option	Meaning	Example
--job-name / -J	Job name	#SBATCH --job-name=rnaseq
--partition / -p	Queue/partition	#SBATCH --partition=batch
--nodes / -N	Number of nodes	#SBATCH --nodes=1
--ntasks / -n	Number of tasks/processes	#SBATCH --ntasks=8
--cpus-per-task / -c	CPU cores per task	#SBATCH --cpus-per-task=4
--mem	Total memory per node	#SBATCH --mem=32G
--gres=gpu:N	Request GPUs	#SBATCH --gres=gpu:1
--time=DD-HH:MM:SS	Days-hours-min-sec	#SBATCH --time=2-00:00:00
--mail-type	Email events	#SBATCH --mail-type=END,FAIL
--output / -o	Standard output log	#SBATCH --output=%x_%j.out
--error / -e	Standard error log	#SBATCH --error=%x_%j.err



Interactive Job

Login nodes have limited CPU and memory resources and should not be used to run jobs.

Please request the appropriate resources and run your jobs on compute nodes using an interactive or batch job.

To start an interactive job:

```
$ srun -p batch -t 2:00:00 -N 1 -n 1 -c 2 --mem=8G --pty bash
```

More details can be found here:

<https://rtguides.it.tufts.edu/hpc/slurm/interactive.html>

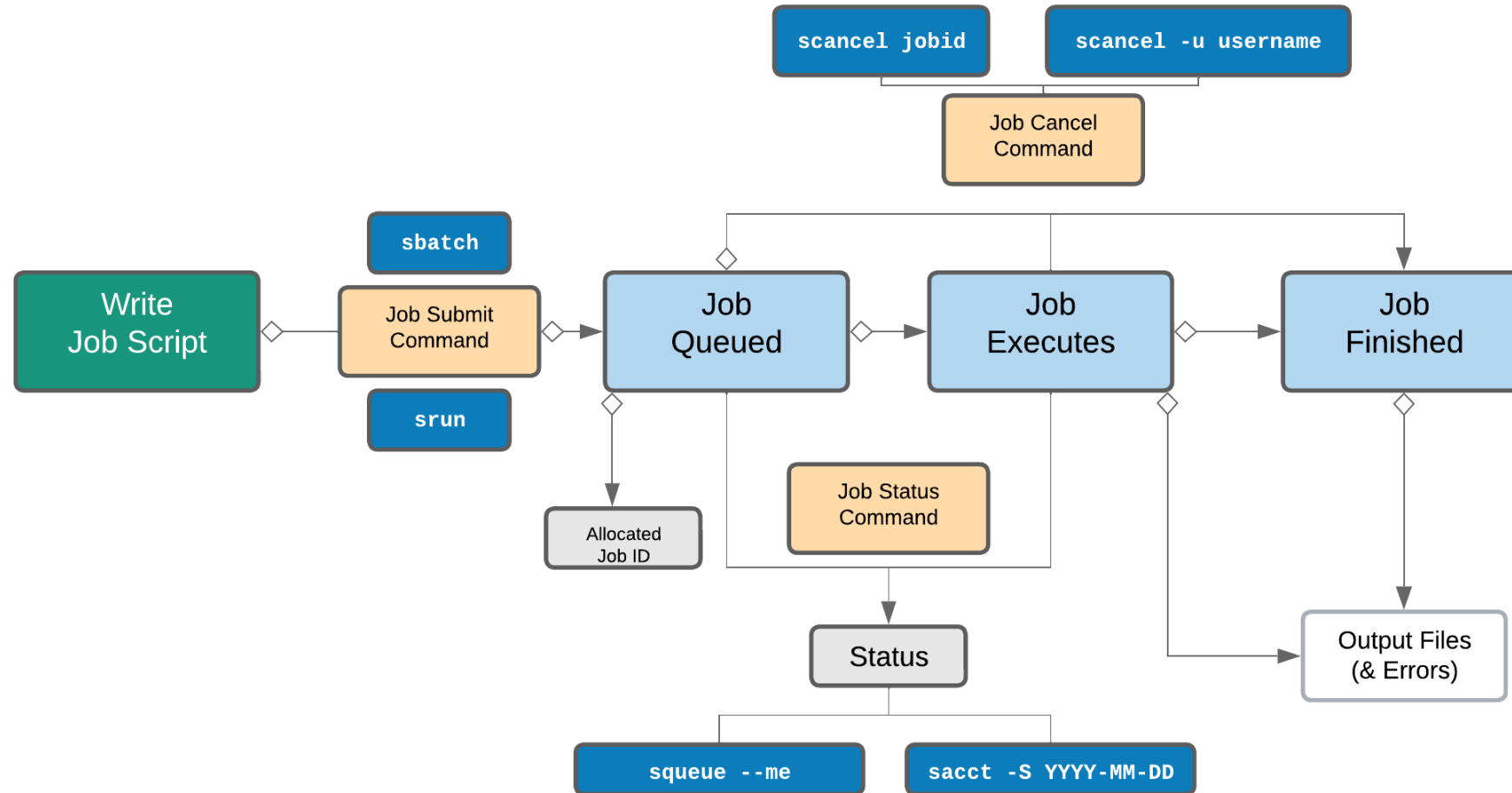


Key SLURM Commands

sbatch script.sh	Submit a batch job
squeue -u \$USER	Show your queued and running jobs
scancel JOBID	Cancel a job
sacct -j JOBID	Check what happened to a completed job
srun -p batch	Interactive compute session
sinfo	Show available partitions and node status



Life cycle of a slurm job



Slurm job script

```
#!/bin/bash
#SBATCH -p batch # Partition to submit to, batch is the standard CPU partition
#SBATCH -N 1 # Number of nodes requested (1 node)
#SBATCH -n 1 # Number of tasks (processes)
#SBATCH -c 8 # Number of CPU cores per task (8 cores total here)
#SBATCH --mem 24G # Total memory allocated for the job (24 GB RAM)
#SBATCH -t 24:00:00 # Wall-clock time limit (HH:MM:SS): job killed after 24 hours
#SBATCH -J my_job_name # Job name (shows in queue; used as %x in log filenames)
#SBATCH --error=%x-%J.err # stderr log file: %x=job name, %J=job ID
#SBATCH --output=%x-%J.out # stdout log file – %x=job name, %J=job ID
```

```
module load nextflow
nextflow run nf-core/rnaseq -r 3.26.0 -profile tufts \
    --input samplesheet.csv \
    --outdir results_rnaseq_test \
    --aligner star_salmon --gtf genes.gtf --fasta genome.fasta
```

```
sbatch rnaseq_script.sh # submit the job to slurm
```



Compose a script with Open Composer

[Home](#) [Application](#) [History](#) [Home Directory](#) [Open OnDemand](#)

Job Script (CPU)

This application generates a job script using only CPU.

Script location*

Select Path

Script name*

Job name

Partition

Number of cores (1 - 64)

Memory (up to 502GB)

Max run time hours (0 - 48)

Max run minutes (0 - 59)

☐ Mail Notifications

☐ Array Job

Script Content

```
#!/bin/bash
#SBATCH -p batch
#SBATCH -N 1
#SBATCH -n 1
#SBATCH --mem 4G
#SBATCH -t 1:0:00
#SBATCH --error=%x-%J-%u.err
#SBATCH --output=%x-%J-%u.out

set -euo pipefail

export OMP_NUM_THREADS="${SLURM_CPUS_PER_TASK:-1}"
# Add your CPU workload below.
# Example:
# module load <module>
# python script.py
```

Submit

Open Composer version: 2.0.0

Modules and Software Applications

<https://rtguides.it.tufts.edu/hpc/application/index.html>

Software Applications

The research computing team strives to make current versions of most common application available for users to run. Applications installed globally on Tufts HPC cluster are managed by environment modules - [Lmod](#).

If you dont see a common application used in your scientific domain please reach out to us about having it installed.

[Suggest Edits](#)

[Show Source](#)

[Report Error](#)

Attention

Some of the software available on the HPC Cluster is not **NOT FREE/OPEN SOURCE SOFTWARE**. Compliance with applicable license agreements is the responsibility of the users running the software on the cluster, failure to comply with license agreements is a violation of our Acceptable Use Policy.

Modules

How to use Lmod on the Tufts HPC cluster to access installed applications.

Conda Environments

How to create conda environment on Tufts HPC cluster.

R

How to run R on Tufts HPC cluster.



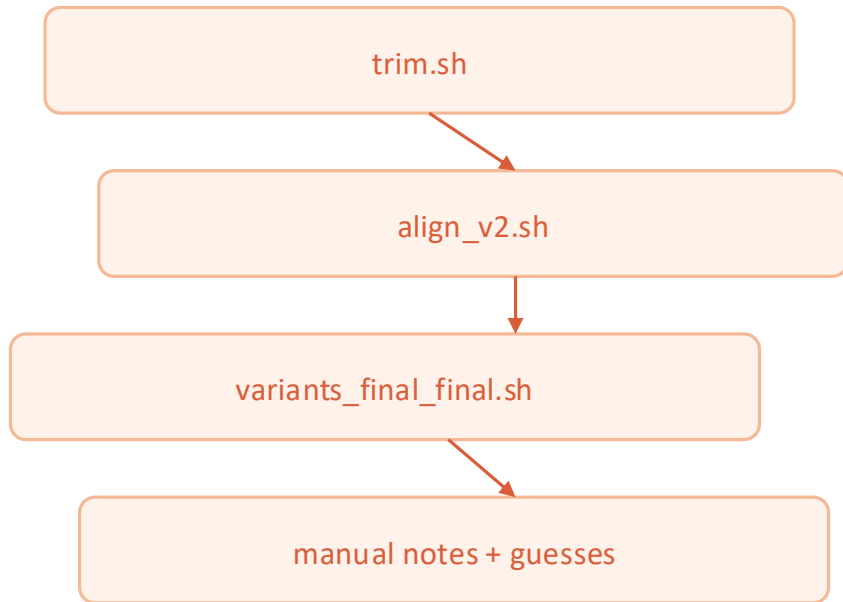
Part2 - Nextflow & nf-core



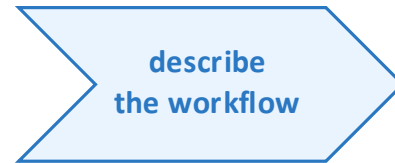
Why biologist need workflow managers

They turn fragile command histories into repeatable, shareable analyses.

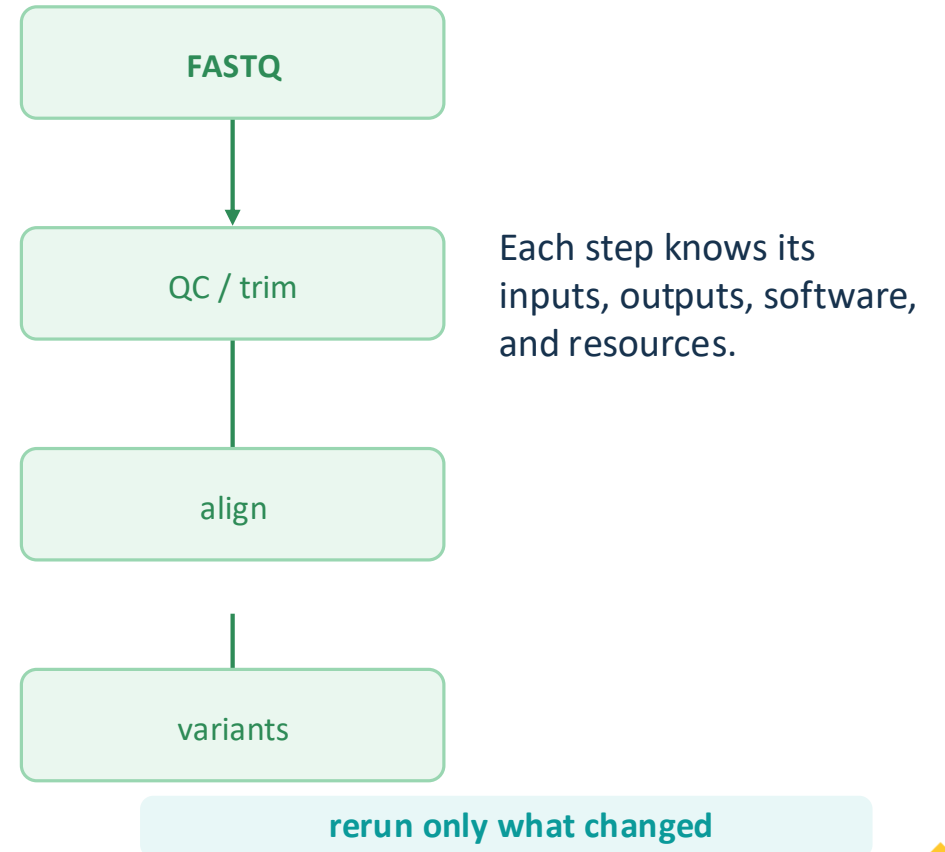
Without a workflow manager



Hard to restart • hard to share • easy to miss a step



With a workflow manager



What is nextflow?

A scripting language built specifically for connecting analysis steps

A workflow manager designed for bioinformatics: you describe the steps and the dependencies, and Nextflow figures out what to run in parallel, where to run it, and how to restart on failure.

1

Parallel by default

Each sample, each chunk, each task runs concurrently — no loop required. Nextflow schedules them for you.

2

Portable executors

Swap local → SLURM → AWS Batch by changing one config line. The pipeline code is untouched.

3

Reproducible by design

Containers (Docker / Singularity) and Conda environments are first-class. Every run is tracked.



Hello, nextflow

Run your first pipeline

```
# Load module
module load nextflow

# Built-in example – nothing to download
nextflow run hello

# Your own hello world:
nextflow run hello.nf --greeting "Hi from HPC"

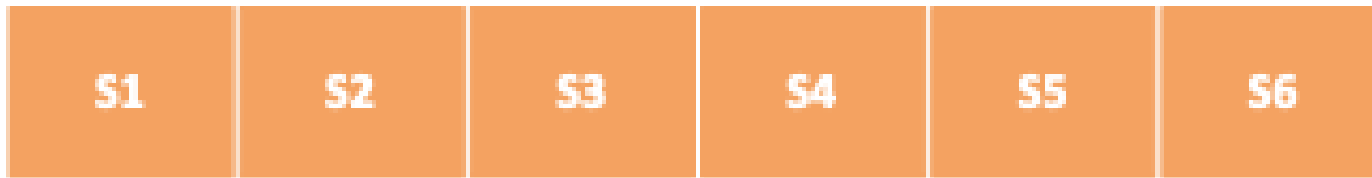
# Use a real community pipeline
nextflow run nf-core/demo -r 1.1.0 -profile test,singularity --outdir demo_output
```



-resume · the killer feature

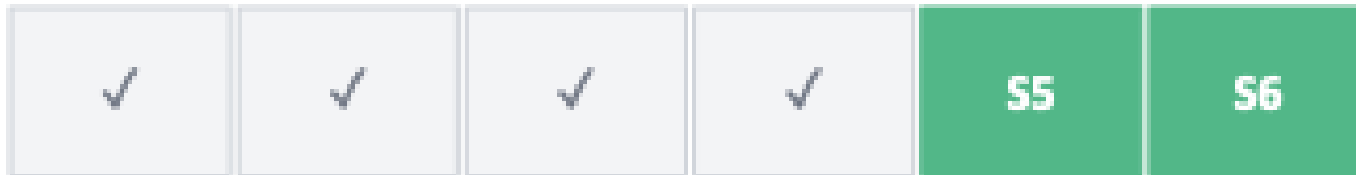
Pipeline crashed at step 5 of 6? Fix the bug. Rerun with -resume.

Without -resume



→ rerun everything

With -resume



→ pick up where you left off

```
nextflow run main.nf -profile slurm -resume
```

Nextflow caches every task in work/ by a hash of its inputs + script. Unchanged tasks are reused.



What is nf-core?

A community, a standard, and a pipeline catalog — all in one

nf-core is a community-driven project that provides a curated set of open-source, peer-reviewed bioinformatics pipelines built on Nextflow. Every pipeline follows the same standards for code, documentation, testing, and reproducibility.

150+

curated pipelines

13,000+

community members

100%

open source

nf-co.re • github.com/nf-core • Slack with 14k+ researchers



The nf-core catalog

Some of the most-used pipelines — chances are your analysis is here

Bulk sequencing	Single cell	Other omics
nf-core/rnaseq Bulk RNA-seq ★	nf-core/scrnaseq Single-cell RNA-seq	nf-core/ampliseq 16S / amplicon
nf-core/sarek Germline + somatic variants	nf-core/scnanoseq Single-cell nanopore	nf-core/mag Metagenomics
nf-core/chipseq ChIP-seq		nf-core/methylseq Bisulfite / methylation
nf-core/atacseq ATAC-seq		nf-core/proteinfold AlphaFold2 / ColabFold

Browse all 150+ at nf-co.re/pipelines — search by keyword, organism, or data type



Three ways to launch

Pick whichever fits your workflow — all produce the same run

nextflow run

what you already know

```
nextflow run nf-  
core/rnaseq \  
  -profile  
test,singularity \  
  --outdir results
```

Direct, scriptable. What we'll use today.

nf-core launch

interactive CLI wizard

```
module load nf-core  
nf-core pipelines launch \  
  rnaseq -r 3.25.0
```

Prompts you through every parameter. Great for first-time use.

Open OnDemand

web UI (tufts specific)

The screenshot shows the Open OnDemand web interface. On the left is a sidebar with a list of 'Interactive Apps' including AI, AI Image Generator, ComfyUI, LLM Notebooks, Research Chatbot, text-generation-webui, Admin tools, Research Technology Guides, Apps, Comsol, Fiji, OpenDroneMap(ODM), Paraview, TensorBoard, Bioinformatics, AlphaFold, and AlphaFold 3. The main panel displays the 'rnaseq 3.25.0' app. It includes a description: 'Launch the nf-core pipeline nf-core-rnaseq version 3.25.0 through Open OnDemand.' Below this is a workflow diagram. Further down, there's a 'Saved form values' section with a 'cache reset utility' button. A 'Number of hours' section has a dropdown menu set to '8' with the text 'Requested waittime for the launch job.' Below that, a 'Which Nextflow executor should be used?' section has a dropdown menu set to 'slurm' with explanatory text: 'With slurm, the pipeline uses its configured scheduler profile. With local, tasks run on the submitted node using a generated config.'

GUI, monitoring, dashboards

nf-core/rnaseq

The most-used bulk RNA-seq pipeline in biology

The standard
end-to-end bulk
RNA-seq pipeline

In one command, you get:

- QC of raw reads (FastQC)
- Adapter & quality trimming
- Alignment (STAR, HISAT2, or Salmon)
- Quantification (gene + transcript)
- Strandedness inference
- Duplicate marking + rRNA filtering
- Gene-level count matrix
- Unified MultiQC report

rnaseq ✓

☆ 1305

RNA sequencing analysis pipeline using STAR, RSEM, HISAT2 or Salmon with gene/isoform counts and extensive quality control.

rna

rna-seq

3.26.0 released 27 days ago

The samplesheet

This one file tells the pipeline what to analyze

```
sample,fastq_1,fastq_2,strandedness
CTRL_REP1,/scratch/$USER/data/ctrl_r1_R1.fq.gz,/scratch/$USER/data/ctrl_r1_R2.fq.gz,auto
CTRL_REP2,/scratch/$USER/data/ctrl_r2_R1.fq.gz,/scratch/$USER/data/ctrl_r2_R2.fq.gz,auto
CTRL_REP3,/scratch/$USER/data/ctrl_r3_R1.fq.gz,/scratch/$USER/data/ctrl_r3_R2.fq.gz,auto
TREAT_REP1,/scratch/$USER/data/treat_r1_R1.fq.gz,/scratch/$USER/data/treat_r1_R2.fq.gz,auto
TREAT_REP2,/scratch/$USER/data/treat_r2_R1.fq.gz,/scratch/$USER/data/treat_r2_R2.fq.gz,auto
TREAT_REP3,/scratch/$USER/data/treat_r3_R1.fq.gz,/scratch/$USER/data/treat_r3_R2.fq.gz,auto
```

sample

Free-form ID. Repeat the same ID if you have multiple lanes — they'll be merged.

fastq_1 / fastq_2

Absolute paths (or URLs) to R1 / R2 FASTQ files. Leave fastq_2 blank for single-end.

strandedness

auto (recommended) — or forward / reverse / unstranded if you know it.



The parameters you'll actually set

Full schema at nf-co.re/rnaseq/parameters — here are the essentials

--input	Path to samplesheet.csv (above)	<i>required</i>
--outdir	Where results go	<i>required</i>
--genome	Reference shortcut (GRCh38, GRCm39, ...) — auto-downloads	<i>Do not use</i>
--fasta / --gtf	OR your own reference FASTA + annotation — skip --genome	
--aligner	star_salmon (default) star_rsem hisat2 bowtie2_salmon	
--skip_<step>	e.g. --skip_markduplicates to speed up / save space	
-profile	Environment: test, singularity, tufts	

--genome uses the genome annotations in iGenomes, which are extremely out of date. For this reason, its use is not recommended.



Putting it all together

One command submits every step to SLURM

```
## -profile tufts can also be used
nextflow run nf-core/rnaseq \
  -profile singularity \
  --input samplesheet.csv \
  --outdir results_rnaseq \
  --fasta genome.fa \
  --gtf genes.gtf \
  --aligner star_salmon \
  -resume
```

-profile singularity

use containers (reproducibility)

-profile tufts

profile designed for tufts hpc

--fasta genome.fa

genome sequence in fasta format

--gtf genes.gtf

genome annotation in gtf format

-resume

skip cached tasks on re-run

For real runs: wrap this in a small sbatch script. Don't run long pipelines on a login node.

The MultiQC report · your first stop

One HTML file, every QC metric. Open it locally in a browser

General Statistics

Per-sample table — %Dups, %GC, M reads, % aligned, M assigned.

FastQC

Per-base quality, adapter content, duplication, overrepresented sequences.

STAR alignment

Uniquely-mapped %, multi-mapped, unmapped, mismatch rate.

Salmon / Qualimap

Quantification summary, gene body coverage, rRNA fraction.

Red flags to look for

- ⚠ < 70% unique alignment rate
- ⚠ High rRNA % (> 10% for polyA-seq)
- ⚠ Very uneven read counts per sample
- ⚠ Obvious adapter contamination post-trim
- ⚠ 5'/3' coverage bias (RNA degradation)
- ⚠ Unexpected strandedness inference

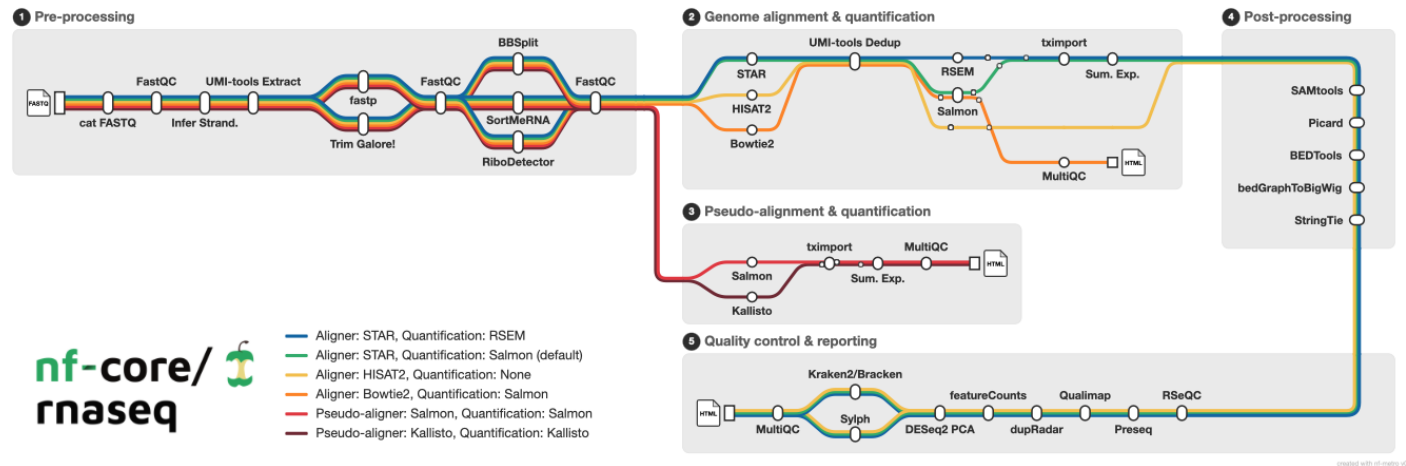


rnaseq in a browser

Same pipeline. Same SLURM. Zero terminal.

rnaseq 3.26.0

Launch the nf-core pipeline [nf-core-rnaseq](#) version 3.26.0 through Open OnDemand.



<https://ondemand-prod.pax.tufts.edu>

Data used in Demo

Solomon et al. *BMC Molecular and Cell Biology* (2021) 22:13
<https://doi.org/10.1186/s12860-021-00343-z>

BMC Molecular and
Cell Biology

RESEARCH ARTICLE

Open Access

Global transcriptome profile of the developmental principles of in vitro iPSC-to-motor neuron differentiation



Emilia Solomon^{1†}, Katie Davis-Anderson^{1†}, Blake Hovde¹, Sofiya Micheva-Viteva^{1*} , Jennifer Foster Harris¹, Scott Twary¹ and Rashi Iyer^{2*}

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